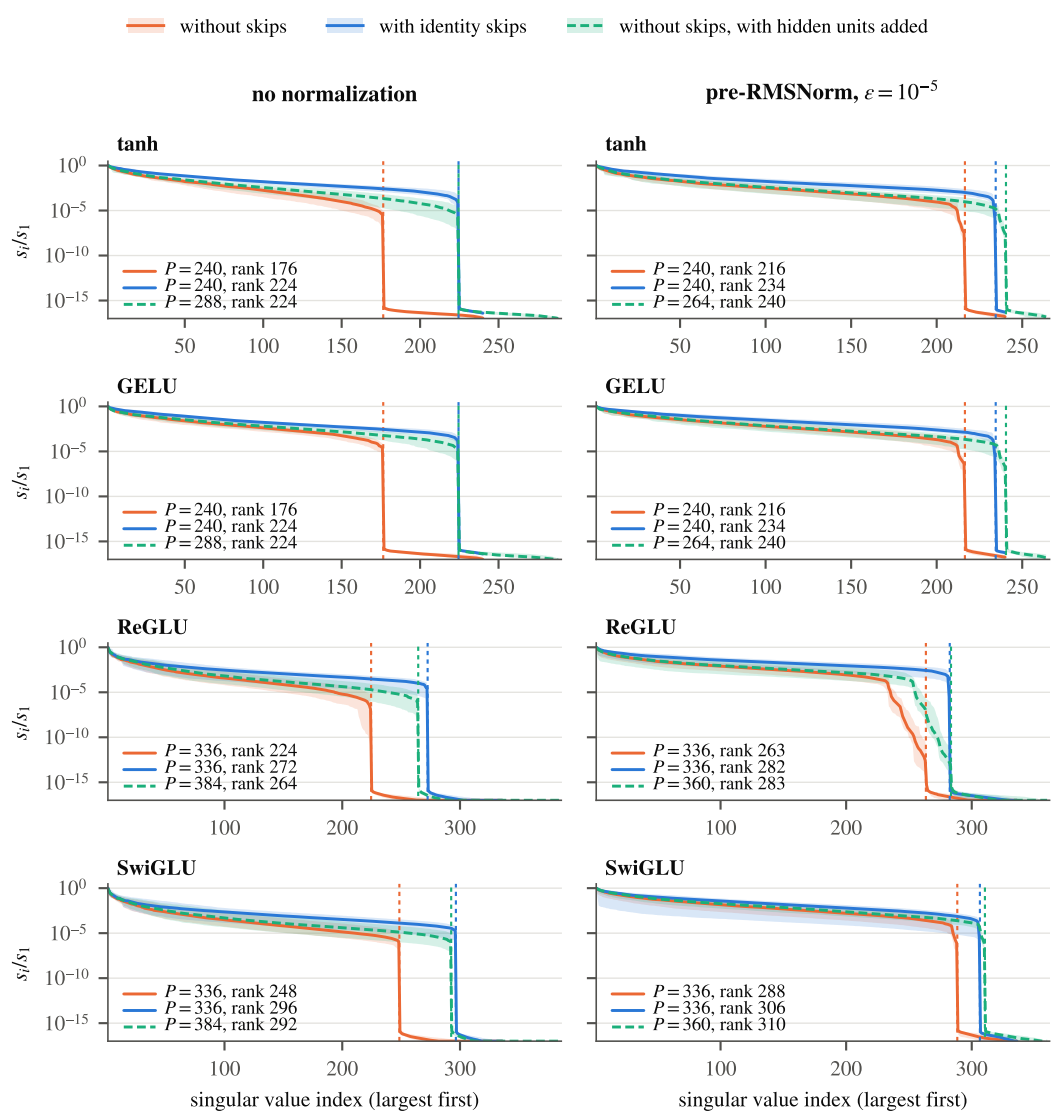

without skips
with identity skips
without skips, with hidden units added
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
tanh
GELU
ReGLU
SwiGLU
s_i/s_1
$P = 240$, rank 176
$P = 240$, rank 224
$P = 288$, rank 224
$P = 240$, rank 216
$P = 240$, rank 234
$P = 264$, rank 240
$P = 336$, rank 224
$P = 336$, rank 272
$P = 384$, rank 264
$P = 336$, rank 263
$P = 336$, rank 282
$P = 360$, rank 283
$P = 336$, rank 248
$P = 336$, rank 296
$P = 384$, rank 292
$P = 336$, rank 288
$P = 336$, rank 306
$P = 360$, rank 310
singular value index (largest first)
singular value index (largest first)